

ConfigDB

After the factory builds your tree

Set, get, and path specificity

- Set stores a typed value under a context path plus field name
- Get asks from a component path for a field
- If several sets match, the longest, most specific, path wins
- A set on env dot agent beats a generic set on env for the same field when the getter sits
- Wrong field name means miss even if another field was set nearby
- Sibling agents do not see each other's sets

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

ConfigDB key path

Practice **set** / **get** by **hierarchical path + field**: who can see a config, and which entry wins when paths nest. Starter: set vif at env . agent, get from env . agent . drv.

Starter example: set("env.agent", "vif", uart_vif) then get from env.agent.drv — hit by walking up.

Load starter example

Challenges 0 / 22 cleared

Quiz: key: A ConfigDB lookup is keyed mainly by...

☐ hierarchical path plus field name

☐ only the simulator seed

☐ VCD dump file name

☐ Makefile target order

Show hint

Check

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Idle

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Core ideas

set

Publish a value under a hierarchical path + field name.

get

Ask from a component path; walk parents for the field.

Specificity

Longer matching path beats a parent set.

Miss

Wrong field or sibling branch → get fails.

ConfigDB lab starter

learn_uvm2017 — uvm configdb

Real UVM literacy

```
❯ uvl - module06-uvm-configdb/examples/navigation

# Track A -- real Unix
# real shell session (Track A -- ConfigDB literacy)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_uvm2017/module06-uvm-configdb

$ ls examples/configdb-sketch/
set-get.txt

$ cat examples/configdb-sketch/set-get.txt
UVM ConfigDB sketch (literacy -- set / get / path)

API pattern (offLine UVM):
uvm_config_dbM(T)::set(cntxt, inst_name, field, value)
uvm_config_dbM(T)::get(cntxt, inst_name, field, value)

Teaching triple (browser lab model):
set(path, field, value)
get(from_path, field) → walk up ancestors of from_path

Starter example:
set(env.agent, "vif", uart_vif)
get("env.agent.drv", "vif") → HIT (walk: drv → agent → env ...)

Lookup rules:
1. Field name must match exactly
2. Set path must be an ancestor of get path (or empty top)
3. Among matches, longest path wins (most specific)

Examples:
set env + set env.agent (same field) → agent child sees env.agent value
set env.agent.vif but get is_active → MISS
set on env.agent0, get from env.agent1.drv → MISS (siblings)

Typical uses:
virtual interface handles to drivers/monitors
is_active / is_passive mode
timeouts, verbosity knobs scoped by path

Timing:
set early (test or env build) before downstream get in build/connect

Common mistakes:
typo in field string
set too late (getter already ran)
expecting sibling agent to see another agent's set

$ head -28 ../../learn_uvm2017_sv_verilator/tools/uvm-2017/1800.2-2017-1.0/src/base/uvm_config_db.svh
//
// Copyright 2010-2010 Mentor Graphics Corporation
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// Copyright 2014 Synfore
// Copyright 2010-2010 Synopsys, Inc.
// Copyright 2010-2010 Cadence Design Systems, Inc.
// Copyright 2011-2012 AMD
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// CONDITIONS OF ANY KIND, either express or implied. See
// the License for the specific language governing
// permissions and limitations under the License.
//
```

Real shell — ConfigDB sketch

Pitfalls to watch

- Do not set after the getter already ran in build
- Do not typo the field string, vif and VIF are different keys
- Do not expect sideways visibility
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run get on the starter and explain the winning path without looking
- On real UVM, sketch one set and one get for passing a virtual interface to a driver
- When you are ready, take the short quiz, then continue to agent anatomy in the next module